

EE2213: Introduction to AI -- Schedule for AY25/26 Semester 1

Week No.	Monday (LT4)	Friday (LT7)
Week 1	Lecture 1: Intro to AI - What is AI - Applications of AI - AI agent	Mini Tutorial 1: Intro to AI Lecture 2: Search I - Uninformed search algorithms: breadth-first, depth-first
Week 2	Lecture 3: Search II - Uniform-cost (Dijkstra's algorithm) - Informed search algorithms: greedy best-first	Tutorial 2: L2-L3 (Release of Assignment 1 (4%))
Week 3	Lecture 4: Search II - Informed search algorithms: A*	Tutorial 3: L4 In-Class Quiz 1 (5%, L1 to L4) (Release of Assignment 2 (6%))
Week 4	Lecture 5: Optimisation I - Linear programming & application	Tutorial 4: L5 (Due of Assignment 1)
Week 5	Lecture 6: Optimisation II - Convex problems - Gradient descent	Tutorial 5: L6 In-Class Quiz 2 (5%, L5 to L6) (Release of Assignment 3 (5%))
Week 6	Lecture 7: Intro to ML and Review of Fundamentals - Definition of ML - Types of ML - Workflow of ML - Fundamentals	Lecture 8: Regression - Linear Regression - Polynomial Regression - Ridge Regression - Evaluation Metrics (Due of Assignment 2)
Recess Week		(Due of Assignment 3)
Week 7	Lecture 9: Classification - Polynomial for Classification - Binomial Logistic Regression - Multinomial Logistic Regression - Evaluation Metrics (Release of Project (20%))	Tutorial 6: L7-L9
Week 8	Lecture 10: Clustering - K-means Clustering - Fuzzy C-means Clustering	Lecture 11: Neural Networks - Perceptron - Multilayer Perceptron - MLP for Image Classification
Week 9	Tutorial 7: L10-L11	In-Class Quiz 3 (5%, L7-L11)
Week 10	Lecture 12: Intro to KRR - Definition of KRR - Types of KR - Types of Reasoning - Case Study: The Wumpus World	Lecture 13: Propositional Logic - Introduction and motivation - Syntax - Semantics - Inference Algorithms (Due of Project) (Release of Assignment 4 (5%))
Week 11	Tutorial 8: L12-L13	In-Class Quiz 4 (5%, L12-L13)
Week 12	Lecture 14: Ethical considerations - Bias in AI - Privacy concerns - Societal impact	Tutorial 14: L14 Start of revision (Due of Assignment 4)
Week 13	Revision Session 1	Revision Session 2

- **Assignment 1 (4%) (2 weeks): Search I (distance calculation).**
- **Assignment 2 (6%) (3 weeks): Search II (A* search).**
- **Assignment 3 (5%) (2 weeks): Optimisation (LP and convex optimization).**
- **Assignment 4 (5%) (2 weeks): Propositional Logic**
- **Project 1 (20%) (4 weeks): Multinomial Logistic Regression for multi-class (>2)**

Optional Online Python Clinic – Session 1 for EE2213:

- **Slot 1: Thursday, 21 Aug, 9:00–10:00 PM**
- **Slot 2: Monday, 25 Aug, 3:00–4:00 PM**

Optional Online Python Clinic – Session 2 for EE2213:

- **Slot 1: Monday, 29 Sep, 3:00–4:00 PM**

Reference book:

Our main reference book is *"Artificial Intelligence: A Modern Approach"* by Russell and Norvig. You can access the e-book through the NUS Library at:
<https://linc.nus.edu.sg/record=b4500759>.